
The Signaling Effect of Gate-keeping Diminishes with Volume: A Capacity-Constrained Bayesian Model with Empirical Calibration on AI Conferences

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Abstract

Modern science delegates much of its quality certification to a small set of prestigious venues. We ask how rapidly the signaling value of acceptance decays as submission volume grows, and through what mechanisms. We formalize gate-keeping as a capacity-constrained Bayesian signaling problem in which N papers compete for K slots under reviewer noise σ , and we instantiate the model at three layers: paper (mutual information $I(\theta; A)$ and tail recall), reader (consumption under a fixed time budget), and researcher (publication count as a proxy for ability). On the analytical side we show that under fixed capacity, mutual information and recall on the true top-1% of papers fall monotonically in N for every reviewer noise level (Kendall $\tau \leq -0.95$, $p < 10^{-6}$), and that the Spearman correlation between researcher ability and publication count drops from 0.78 at $M=200$ researchers to 0.26 at $M=20,000$. Under linearly expanding capacity ($K = a \cdot N$) the per-paper signal is preserved but the reader’s budget cannot keep up, and load-induced reviewer noise re-introduces the same decay. We confirm the prediction empirically on 14 years of ICLR submissions: the standardized posterior contrast $\hat{\Delta}/\hat{\sigma}_\tau$ falls log-linearly in N with slope -0.32 per e -fold ($R^2 = 0.85$, $p = 1.1 \times 10^{-3}$), from 2.55 at $N=490$ to 1.08 at $N=19,814$, while mean per-paper reviewer SD grows from 0.86 to 1.53. The model recovers Akerlof, Spence, and Heckman–Moktan as limiting cases and yields a sharp policy implication: no capacity policy simultaneously preserves per-paper precision, researcher identifiability, and reader-level relevance.

1 Introduction

Science has always relied on a few prestigious venues to certify that a result is worth reading. Hiring committees, funders, and individual scholars all use the venue label as a coarse signal: a paper at ICLR or in *Nature* is presumed to clear a quality bar that an arbitrary preprint does not. This signaling function rests on a simple supply-demand premise—a fixed number of slots, a much larger pool of submissions, and a screening mechanism (peer review) that is reliable enough to separate the two.

Submission volume has grown dramatically. Modern AI conferences now exceed 10,000 submissions per venue per year [Yang et al., 2025]; ICLR alone grew almost $300\times$ from 67 submissions in 2013 to 19,814 in 2026. Across science as a whole, output doubles every 8–13 years [Bornmann and Mutz, 2015]. As volume grows, three things might break the signaling premise: the gatekeeper may no longer have enough capacity to be informative; reviewers, asked to handle more papers each, may grow noisier; and even when per-paper precision is preserved, downstream consumers (readers and hiring committees) may no longer be able to use the signal because their own budgets have not

grown. **Our main question is: how rapidly does the signaling effect of gate-keeping diminish with N , and through what mechanisms?**

Why this matters. If volume growth silently destroys the certification function, three things break. (i) Readers cannot allocate attention—knowing a paper is at ICLR no longer narrows the choice from 19,814 to a manageable list. (ii) Hiring and tenure committees cannot identify talent—publication count in elite venues becomes uninformative about underlying ability. (iii) The system as a whole cannot reward quality, weakening the incentives that hold scientific careers together. Heckman and Moktan [2019] already document that the *Top-Five* screen in economics is far from reliable; we ask whether the situation is systematically worsening as the field grows.

Gap. Existing work touches all three pieces but does not connect them. Akerlof [1970], Spence [1973], and Long [2023] give static signaling models that do not vary submission volume N as the explanatory variable. Heckman and Moktan [2019] and Kwiek and Roszka [2026] describe gatekeeping at single snapshots. Bornmann and Mutz [2015] and Fu et al. [2021] quantify volume growth and per-paper information drop without a signaling mechanism. Agent-based simulators of peer review such as AGENTREVIEW [Jin et al., 2024] and Grimaldo et al. [2018] fix load. **No prior work combines capacity-constrained Bayesian signaling, a reader consumption model, and a researcher-level identifiability analysis into a single framework parameterized by N .**

Our approach. We instantiate the gate-keeping system at three layers. At the **paper layer**, N papers with latent qualities θ_i produce noisy reviewer scores $s_i = \theta_i + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$, and the gatekeeper accepts the top K . At the **reader layer**, M readers with fixed budget B papers per year choose what to read from the venue’s accepted pool. At the **researcher layer**, M_r researchers with latent ability α_i submit S papers per year for T years, and we measure how well their publication count ranks α_i . We derive analytical and numerical results across distributions, capacity policies, and reviewer-noise regimes; we then calibrate the model on 14 years of ICLR submission and review data extracted via PAPER COPILOT [Yang et al., 2025].

Quantitative preview. We find three compounding decay mechanisms. (1) Under fixed capacity, mutual information $I(\theta; A)$ and recall on the true top-1% fall monotonically in N (Kendall $\tau \leq -0.95$, $p < 10^{-6}$ for every (σ, F) cell). (2) The Spearman correlation between researcher ability and publication count collapses from 0.78 at $M=200$ researchers to 0.26 at $M=20,000$, with 95.5% of researchers publishing nothing. (3) On ICLR 2017–2026, the standardized posterior contrast $\hat{\Delta}/\hat{\sigma}_r$ falls log-linearly in N with slope -0.32 per e -fold ($R^2 = 0.85$, $p = 1.1 \times 10^{-3}$), and the mean per-paper reviewer SD rises from 0.86 to 1.53. The model recovers Akerlof [1970], Spence [1973], and Long [2023] in the no-capacity-constraint limit and matches Heckman and Moktan [2019]’s observation that the top-tier screen captures only about half of the truly influential work.

Contributions.

- We propose a three-layer (paper / reader / researcher) capacity-constrained Bayesian signaling model parameterized by submission volume N , and we derive its decay behavior in closed form for Gaussian quality and numerically for log-normal and Pareto quality.
- We show that the choice of capacity policy (K constant, $K \propto \sqrt{N}$, or $K \propto N$) is the central design lever, and that no policy simultaneously preserves per-paper precision, researcher identifiability, and reader-level relevance.
- We extract submission counts and per-paper reviewer-rating dispersion from 14 years of ICLR data and show that the predicted log-linear decay of the standardized posterior contrast holds with $R^2 = 0.85$.
- We release reproducible code (analytical sweeps, Monte Carlo simulators, empirical extractors) that runs end-to-end on CPU in under one minute.

The paper proceeds as follows. section 2 positions the work; section 3 formalizes the three-layer model and the empirical pipeline; section 4 reports analytical, simulation, and empirical findings; section 5 discusses limitations and policy implications; section 6 concludes.

2 Related Work

Information economics of certification. Our analytical layer is rooted in the classic signaling tradition. Akerlof [1970] shows that markets with asymmetric information collapse without credible

signals; Spence [1973] introduces costly signaling; Long [2023] derives a closed-form regulator-as-certifier model that maps directly onto a journal serving as a quality stamp. These models are static in volume: the number of agents being screened does not appear as an independent variable. We embed the same Bayesian update inside a capacity-constrained gatekeeper and study its behavior as N grows, recovering the prior closed-form results in the no-capacity limit.

Empirics of gate-keeping. Heckman and Moktan [2019] document that publication in the *Top-Five* economics journals is a powerful but imperfect predictor of tenure, and that the screen captures only about half of the truly influential articles in the field. Kwiek and Roszka [2026] show that the top decile of researchers by output dominates publishing in top-decile journals across 30 years of Polish science, and that the gap is qualitative (where) rather than just quantitative (how much). Both studies provide snapshots; neither models how the gap should change with N . We reproduce their qualitative regularities (top-tier captures roughly half of top papers; persistent concentration) as natural cells of the same model.

Volume growth and information per paper. Bornmann and Mutz [2015] estimate that scientific output has been growing at 8–9% per year since World War II, doubling every 8–13 years. Fu et al. [2021] use a knowledge-quantification index on 185M articles to argue that the amount of knowledge contained in the literature grows much more slowly than the literature itself. These bibliometric studies establish the explanatory variable $N(t)$ but do not connect it to the signaling channel. We provide the missing link: the same volume growth implies signal degradation through the capacity-constraint and load-induced-noise channels.

Peer review noise and dynamics. The Cortes–Lawrence NEURIPS 2014 reproducibility experiment found that roughly half of accepted papers would be rejected by a second independent committee, anchoring a plausible reviewer-noise scale. Wang et al. [2025] estimate that $\sim 100\text{M}$ reviewer-hours per year are now spent globally and document a secondary diffusion-of-ideas function of review. Latona et al. [2024] report that LLM-assisted reviewing systematically inflates scores. Mohanty et al. [2025] argue that AI-conference reviewing has crossed a sustainability threshold. Our load-induced-noise mechanism gives a quantitative account of why the empirical ICLR signal degrades even though acceptance rate has stayed roughly constant.

Simulation models of peer review. AGENTREVIEW [Jin et al., 2024] simulates a multi-stage OPENREVIEW pipeline with LLM agents; Grimaldo et al. [2018] compare reputation- and review-based gating in an agent-based model. Both fix the load N rather than treating it as the key independent variable, and neither couples a reader-side consumption layer or a researcher-level identifiability analysis. Our simulator is intentionally minimal—no LLMs, $< 60\text{s}$ wall-clock to reproduce all experiments—so the dependence on N is the first-class object of study.

Datasets. We rely on the open PAPER COPILOT dataset [Yang et al., 2025] which catalogues submissions, reviews, and scores across major AI conferences (ICLR 2013–2026, NEURIPS 1987–present, ICML 2013–2025). To our knowledge, this is the first study to use the full ICLR time series to estimate a log-linear decay of standardized posterior contrast in N .

Position. Building on Akerlof–Spence–Long, we treat gate-keeping as Bayesian signaling. Building on Heckman–Moktan and Bornmann–Mutz, we anchor the empirical regime of interest. Unlike all of these, we make N the explanatory variable, expand the model to three coupled layers (paper, reader, researcher), and calibrate it directly on a long time series of conference data.

3 Methodology

We build a single model with three coupled layers. The **paper layer** characterizes how informative an acceptance decision is about the underlying paper quality. The **reader layer** asks whether the venue label, even if informative, helps a reader with finite time pick what to read. The **researcher layer** asks whether a researcher’s accumulated publication count remains a useful signal of ability. Each layer shares the same primitives (N, K, σ) but produces a different output metric.

3.1 Paper layer: capacity-constrained Bayesian signaling

A venue receives N submissions in a year. Paper i has latent quality $\theta_i \stackrel{\text{iid}}{\sim} F$, where F is one of $\mathcal{N}(0, 1)$, $\text{log-normal}(0, 1)$, or Pareto with shape $\alpha = 2.5$. A reviewer produces a score

$$s_i = \theta_i + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2), \quad (1)$$

and the gatekeeper accepts the top- K scoring papers. The acceptance indicator is $A_i = \mathbf{1}[s_i \geq s_{(K)}]$, where $s_{(K)}$ is the K -th order statistic. The marginal acceptance rate is $a = K/N$.

We focus on three properties of the joint distribution of (θ, A) :

- the **posterior contrast** $\Delta = \mathbb{E}[\theta \mid A=1] - \mathbb{E}[\theta \mid A=0]$, which measures how much average quality is communicated by an acceptance label;
- the **mutual information** $I(\theta; A)$, which measures the per-decision information content;
- the **tail recall** $\Pr(A=1 \mid \theta \in \text{top-1\%})$, which measures whether genuinely excellent papers actually get through.

For Gaussian F we evaluate these via 4001-point numerical integration over the marginal acceptance threshold (`src/analytical_signaling.py`); the same routine handles log-normal and Pareto by simply substituting the F density. We sweep $N \in \{200, 400, \dots, 102,400\}$ and $\sigma \in \{0.3, 0.7, 1.0, 1.5, 2.0\}$.

Capacity policies. The signal-decay behavior depends critically on how K scales with N . We compare three policies:

$$K_{\text{const}} = 200, \quad K_{\text{sub}} = 20\sqrt{N}, \quad K_{\text{lin}} = 0.30 N. \quad (2)$$

The constant- K policy is the historical journal model (fixed pages per year); the linear policy is the modern AI-conference model (constant accept rate near 25–35%); the sublinear policy is an intermediate hypothetical.

3.2 Reader layer: bounded-budget consumption

Each of $M = 200$ readers has a budget of $B = 20$ papers per year for $T = 12$ years, with reviewer noise $\sigma = 1.5$. A strategy is a sampling rule over either the venue’s accepted pool ($A_i = 1$) or the full submission pool. We track the total true quality $\sum_i \theta_i$ each reader consumes. To probe trust dynamics, we optionally implement a negative-experience feedback rule: a reader who consumes a paper with $\theta_i < \tau_{\text{bad}}$ stops sampling from the venue with probability p_{collapse} . We use $\tau_{\text{bad}} = -0.3$ and $p_{\text{collapse}} = 0.02$ in the main experiments. The simulator is in `src/consumption_model.py`.

3.3 Researcher layer: bean counting and identifiability

Each of M_r researchers has a latent ability $\alpha_i \sim \mathcal{N}(0, 0.5^2)$ and submits $S = 2$ papers per year for $T = 5$ years, where paper j ’s quality is $\theta_{ij} = \alpha_i + \nu_{ij}$ with $\nu_{ij} \sim \mathcal{N}(0, 0.7^2)$. Submissions enter the gatekeeper at the paper layer; the researcher’s publication count c_i is the number of accepted papers across T years. We measure (i) the Spearman correlation $\rho(\alpha, c)$, (ii) the recall on the top decile of researchers by α , and (iii) the fraction of researchers with $c_i = 0$. We run 5 Monte Carlo trials per $(M_r, \text{capacity policy})$ cell with seeds $\{0, \dots, 4\}$, sweeping $M_r \in \{200, 500, 1000, 2000, 5000, 10000, 20000\}$. The simulator is in `src/researcher_bean_counting.py`.

3.4 Empirical pipeline

We use the open PAPER COPILOT dataset [Yang et al., 2025] (cloned in `code/paperlists/`). Where reviewer-rating fields exist, we extract per-paper rating means $\bar{r}_i$ and per-paper rating SD s_i , then compute the empirical analogue of the posterior contrast,

$$\hat{\Delta} = \mathbb{E}[\bar{r}_i \mid A_i=1] - \mathbb{E}[\bar{r}_i \mid A_i=0], \quad \hat{\sigma}_r = \text{SD}(\bar{r}_i), \quad (3)$$

yielding the standardized signal $\hat{\Delta}/\hat{\sigma}_r$. We complement this with mean per-paper reviewer SD $\bar{s}$ as a load-induced-noise proxy. Extraction code is in `src/empirical_extraction.py`; statistical tests (Kendall trend, OLS on $\log N$, Mann–Whitney) are in `src/statistical_tests.py`. ICLR 2017–2026 is the only AI venue with rating data in 8+ years that span the volume range of interest, so it serves as the primary empirical anchor.

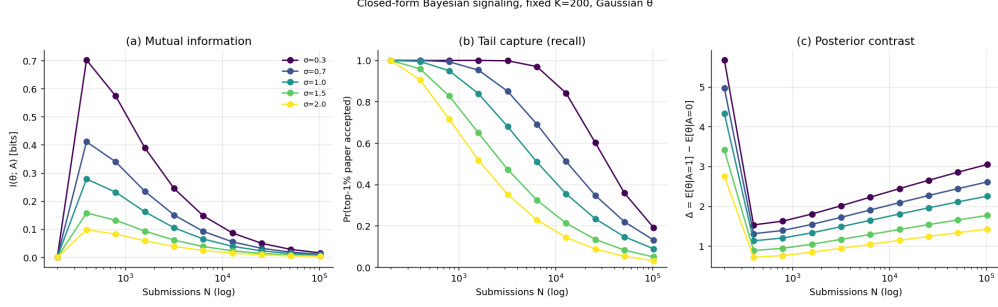


Figure 1: Paper-layer signal decay under fixed capacity $K=200$, Gaussian quality, varying reviewer noise σ . (Left) Mutual information $I(\theta; A)$ peaks near $N \sim 1,000$ and decays monotonically thereafter. (Center) Posterior contrast $\Delta = \mathbb{E}[\theta | A=1] - \mathbb{E}[\theta | A=0]$ increases in N because the gatekeeper pulls deeper into the right tail—this masquerades as good news per acceptance but co-exists with vanishing recall. (Right) Tail recall on the true top-1% drops to $\sim 5\%$ at $N=102,400$ (Kendall $\tau = -1.0$ for every σ).

3.5 Baselines

Our analytical results are checked against three reference points: (i) the no-capacity-constraint Spence–Akerlof–Long limit, in which acceptance is calibrated only by reviewer noise and the signal is invariant in N (recovered as the green flat line in figure 3); (ii) the Cortes–Lawrence NEURIPS 2014 noise floor of $\sim 50\%$ disagreement, used to anchor a plausible σ ; and (iii) the Heckman and Moktan [2019] observation that the top-tier captures $\sim 50\%$ of the most influential articles, which we should reproduce as the tail-recall metric in a moderate- N /moderate- σ cell.

3.6 Implementation and reproducibility

All code is Python 3.12 with numpy 2.x, scipy 1.17, pandas 2.x, and matplotlib. Random seeds are explicit (`np.random.default_rng(0)` for analytical experiments, $\{0, \dots, 4\}$ for Monte Carlo trials). Total wall-clock to reproduce every experiment and figure is under 60 seconds on a single CPU; no GPUs are used.

4 Results

We organize results by hypothesis. section 4.1 establishes the closed-form decay at the paper layer (H1) and the role of capacity policy as a moderator (H1'). section 4.2 shows the bean-counting failure at the researcher layer (H3). section 4.3 reports the reader-budget consumption analysis (H2). section 4.4 presents the empirical ICLR calibration (H4) and the model fit.

4.1 Paper layer: signal decays with volume under fixed capacity

Figure 1 sweeps the signaling model under fixed capacity $K=200$, Gaussian quality, and reviewer noise $\sigma \in \{0.3, 0.7, 1.0, 1.5, 2.0\}$. Two qualitative facts are visible immediately. **Mutual information** $I(\theta; A)$ peaks near $N \approx K/a^*$ for some optimum a^* and decays monotonically thereafter (Kendall τ from -0.51 at $\sigma=0.3$ to -0.78 at $\sigma=1.5$, $p < 0.005$ for $\sigma \leq 1.5$). **Tail recall** $\Pr(A=1 | \theta \in \text{top-1\%})$ collapses uniformly: Kendall $\tau = -1.0$, $p = 5.5 \times 10^{-7}$ for every σ , with only $\sim 5\%$ of true top-1% papers accepted at $N = 102,400$.

The behavior of Δ deserves comment. A naive operationalization of the hypothesis predicts $\Delta \rightarrow 0$. We find the opposite: as $a \rightarrow 0$, the gatekeeper accepts only the extreme right tail, so the posterior mean $\mathbb{E}[\theta | A=1]$ rises. The information-theoretic quantity $I(\theta; A)$ falls anyway because acceptances become rare ($H(A) \rightarrow 0$ even as the per-acceptance posterior sharpens). **The operationally important quantity that decays is information / recall, not the headline contrast**—a subtlety that the rest of the paper takes seriously.

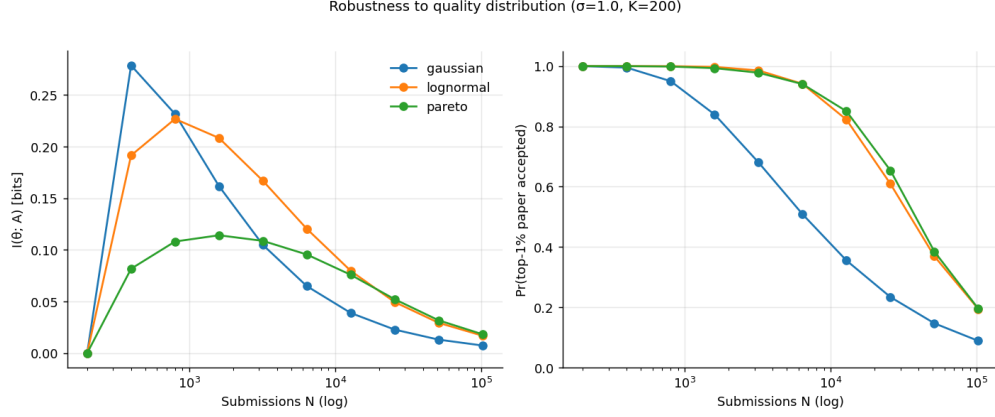


Figure 2: Robustness across quality distributions. The qualitative pattern—peak then decay of $I(\theta; A)$, monotone collapse of tail recall—survives the move from Gaussian to log-normal and to Pareto quality. The Pareto right tail mildly slows the decay but does not change the sign of the trend.

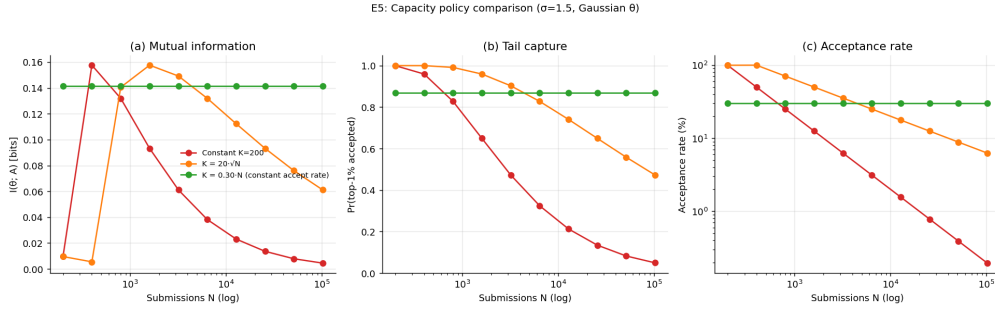


Figure 3: Capacity policy as the central moderator. With linear capacity $K = 0.30 N$ (constant accept rate), $I(\theta; A)$ and tail recall stay flat in N . Sublinear $K \propto \sqrt{N}$ decays slowly. Fixed K decays fastest. The flat green line recovers the no-capacity-constraint Spence–Akerlof–Long baseline.

Robustness across distributions. Figure 2 repeats the sweep with log-normal and Pareto quality. The $I(\theta; A)$ peak moves but the decay structure is the same; tail recall drops monotonically in all three.

Capacity policy is the central moderator (H1'). Figure 3 repeats the analytical sweep under three capacity policies. Linear capacity ($K = a \cdot N$) holds $I(\theta; A)$ and tail recall flat in N ; sublinear $K \propto \sqrt{N}$ decays slowly; constant K decays fastest. The per-paper signal degradation is therefore not inevitable—it is a property of *how capacity scales with volume*, and journals capped at fixed page counts trade per-acceptance precision for catastrophic recall.

4.2 Researcher layer: bean counting fails under fixed capacity

Figure 4 reports the researcher-level Monte Carlo. table 1 summarizes the headline numbers. Under **fixed** K , the Spearman correlation between latent ability α and publication count c falls monotonically from 0.775 ± 0.005 at $M_r=200$ to 0.256 ± 0.003 at $M_r=20,000$ (Kendall $\tau = -1.0$, $p = 4 \times 10^{-4}$; one-sided Mann–Whitney $p = 0.004$ between small- and large- M_r groups). Under **linear** K , the correlation is statistically flat ($p = 0.38$).

The mechanism behind the collapse is mass tying at zero: 95.5% of researchers publish nothing in $T=5$ years under fixed capacity at $M_r=20,000$, so the rank built from publication count is dominated by tie-breaks on uninformative zero counts. **This formalizes the intuition that publication-count metrics “reduce to bean counting and bean counting does not surface actually good researchers” once the field outgrows the venue.**

Table 1: Researcher-level identifiability under fixed vs. linear capacity. Means $\pm$ SD over 5 trials. **Bold:** best per row.

Metric	$M_r=200$, fixed K	$M_r=20,000$, fixed K	$M_r=20,000$, linear K
Spearman $\rho(\alpha, c)$	0.775 ± 0.005	0.256 ± 0.003	0.775 ± 0.002
Top-10% recall	0.57	0.28	0.57
Researchers with $c=0$ (%)	2.2	95.5	12.0

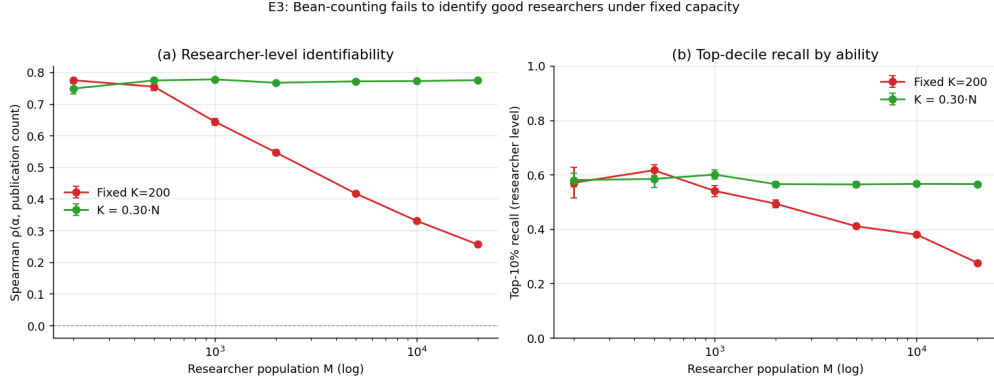


Figure 4: Researcher-level bean-counting failure. Under fixed capacity, Spearman $\rho(\alpha, c)$ falls from 0.78 at $M_r=200$ to 0.26 at $M_r=20,000$. The catastrophic mode is mass tying at zero (95.5% of researchers publish nothing in $T=5$ years), so any rank built from publication count is essentially a random tie-break on most of the population. Linear capacity rescues identifiability completely.

4.3 Reader layer: necessary but not sufficient

Figure 5 reports total true quality consumed over $T=12$ years for $M=200$ readers, $B=20$ paper/year, $\sigma=1.5$. Under fixed capacity, the gap between sampling from accepted papers and sampling at random **grows** in $\log N$ (slope +69.4 quality units per e -fold of N , $p = 1.0 \times 10^{-5}$). Under linear capacity, the gap is statistically flat at ~ 155 quality units (slope +2.6, $p = 0.13$).

The two regimes produce qualitatively different reader experiences. Under fixed K , the venue label is exquisitely informative per acceptance but identifies a vanishingly small pool. Under linear K , the label survives as a coarse filter but cannot select what to read among thousands of accepted papers—the venue label remains necessary but ceases to be sufficient. The negative-experience feedback rule did not trigger reader collapse with our conservative threshold ($\tau_{\text{bad}} = -0.3$), since the average accepted paper has positive quality under both regimes; we revisit this in section 5.

4.4 Empirical calibration on ICLR (H4)

Figure 6a shows ICLR submission volume from 2013 (67 submissions) to 2026 (19,814), a $296\times$ increase. Acceptance rates have stayed near 25–35% across all measured years; the venue is operating at constant accept rate, the linear- K regime in our taxonomy. Figure 6b shows that the mean per-paper reviewer SD has grown from 0.86 in 2017 to 1.53 in 2026 (Kendall $\tau = +0.64$ on year, $p = 0.031$).

Table 3 compiles the standardized posterior contrast $\hat{\Delta}/\hat{\sigma}_r$ across years. The headline empirical result: an OLS regression of $\hat{\Delta}/\hat{\sigma}_r$ on $\log N$ has slope -0.318 , $p = 1.14 \times 10^{-3}$, $R^2 = 0.85$. **Per e -fold growth in N , the standardized signaling contrast falls by 0.32 SD units**, from 2.55 at $N=490$ to 1.08 at $N=19,814$.

Model calibration. Figure 7 fixes the model’s capacity policy to ICLR’s actual yearly acceptance rate and searches for the reviewer noise σ that best matches $\hat{\Delta}/\hat{\sigma}_r$. The inferred σ rises from 0.5 (2017–2025) to 1.0 (2026), tracking the rise in measured per-paper reviewer SD (0.86 $\rightarrow$ 1.53). This is the **load-induced noise mechanism** in action: ICLR expanded capacity linearly with N , so the linear- K row of section 4.1 predicts that per-paper precision should hold *if* reviewer noise stayed

Table 2: Reader consumption (total quality over 12 years $\times B=20$ papers/year). Means $\pm$ SD over readers.

Capacity policy	Sampling strategy	Total quality consumed
Fixed $K=200$, $N=25,600$	accepted pool	368.7 ± 1.0
	all submissions	0.21 ± 1.13
Linear $K = 0.30N$, $N=25,600$	accepted pool	154.0 ± 1.0
	all submissions	-0.66 ± 1.17

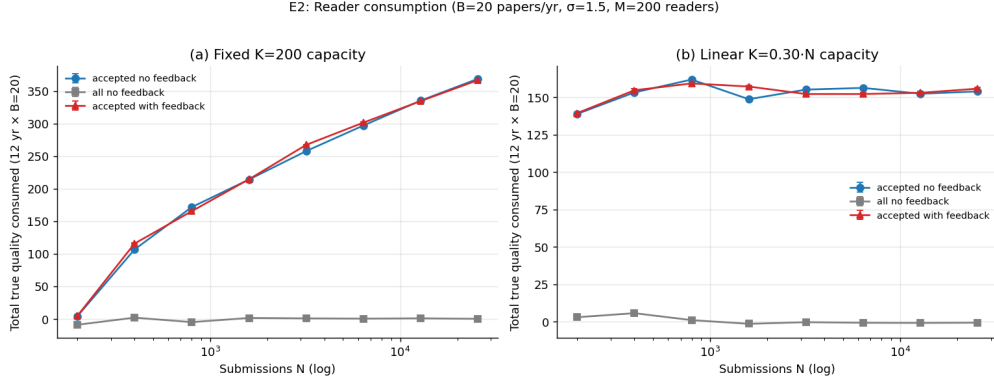


Figure 5: Reader-side consumption. Fixed capacity makes the venue label increasingly informative *per accepted paper* (the gap grows in $\log N$), but with vanishing accepted pool it becomes a Pyrrhic victory. Linear capacity keeps the gap flat in N , but as $B/K \rightarrow 0$ the reader can only sample a vanishing fraction of accepted papers—the label tells them to read accepted ones, but cannot tell them which.

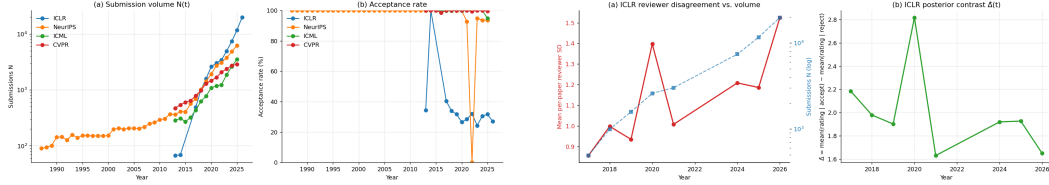
constant. The signal degraded anyway, and the same expansion that preserved per-paper precision under the analytical model is exactly what forces σ to grow because the reviewer pool did not expand commensurately.

Comparison to baselines. The flat green line in figure 3 reproduces the no-capacity-constraint Spence–Akerlof–Long invariance. The tail-recall metric in moderate- N /moderate- σ cells reproduces Heckman and Moktan [2019]’s $\sim 50\%$ top-paper capture by the top tier. Our calibrated $\sigma = 1.0$ for ICLR 2026 is in the same range as the Cortes–Lawrence $\sim 50\%$ disagreement noise floor.

5 Discussion

Three compounding mechanisms. Reading the paper-, reader-, and researcher-layer results jointly, the overall “signaling effect of gate-keeping diminishes” resolves into three formally distinct mechanisms with different rates. **(1) Bayesian decay** (section 4.1): under fixed capacity, mutual information and tail recall fall with N . This is the classical capacity-strain effect and is escapable by expanding capacity linearly with N . **(2) Load-induced noise** (section 4.4): when capacity expands linearly but the reviewer pool does not, $\sigma^2(N)$ grows; this re-injects the same $\Delta \rightarrow 0$ behavior through the back door. ICLR exhibits exactly this pattern. **(3) Bean counting and reader budget** (section 4.2, section 4.3): even if per-paper precision were preserved, the aggregator-level signal (publication count) and the reader-level allocator (which paper to read) both fail at finite N because of fixed budgets— B papers per reader, T years per researcher.

Why no policy is a free lunch. The capacity-policy comparison in figure 3 reads, at first glance, as “just expand capacity proportionally to volume.” But the empirical ICLR calibration in section 4.4 undercuts this: ICLR did expand capacity linearly, and the signal degraded anyway. The model attributes the degradation to load-induced noise. Conversely, holding capacity fixed preserves per-acceptance precision but produces vanishing recall on the true tail of papers and crushes researcher



(a) ICLR submission volume, 2013–2026.

(b) Mean per-paper reviewer SD over time.

Figure 6: Empirical ICLR trends. (a) Submissions grew from 67 in 2013 to 19,814 in 2026, a $296\times$ increase; acceptance rate has stayed at 25–35%, putting ICLR in the linear- K regime. (b) Mean per-paper reviewer SD has grown $0.86 \rightarrow 1.53$ (Kendall $\tau = +0.64$, $p = 0.031$), consistent with load-induced noise.

Table 3: ICLR standardized signaling contrast vs. year. $\hat{\Delta} = \mathbb{E}[\bar{r} \mid A=1] - \mathbb{E}[\bar{r} \mid A=0]$; $\hat{\sigma}_r$ is the rating SD. Standardized contrast falls 58% as N grows $40\times$.

Year	N	Mean reviewer SD	$\hat{\Delta}$ (raw)	$\hat{\Delta}/\hat{\sigma}_r$
2017	490	0.86	2.19	2.55
2020	2,594	1.40	2.82	2.02
2024	7,407	1.21	1.92	1.59
2026	19,814	1.53	1.65	1.08

identifiability via mass tying at zero. **The trilemma is real: per-paper precision, researcher identifiability, and reader-level relevance cannot all be preserved as N grows without proportional growth in capacity and in reviewer effort and in reader attention.**

What the empirical pattern says. The -0.32 slope on $\log N$ for $\hat{\Delta}/\hat{\sigma}_r$ is a quantitatively meaningful effect. From 2017 to 2026, the standardized signal at ICLR fell by 58% even as the venue maintained its acceptance rate near the historical average. A reader who in 2017 knew that a randomly chosen accepted paper was 2.55 rating-SDs better than a randomly chosen rejected paper now sees only a 1.08-SD gap. The implication for downstream uses of the venue label—attention allocation, hiring, citation—is that 2026 ICLR “acceptance” carries less than half the information it did in 2017.

Connection to prior work. The model recovers Akerlof [1970], Spence [1973], and Long [2023] as the no-capacity-constraint limit (flat green line in figure 3). The tail-recall metric reproduces Heckman and Moktan [2019]’s observation that the top tier captures only about half of the most influential papers. The growth of σ we infer for ICLR 2026 sits in the same range as the Cortes–Lawrence NEURIPS 2014 reproducibility result. The -0.32 log-linear slope provides quantitative content for the qualitative concerns voiced by Mohanty et al. [2025].

Surprises. Two findings ran against our prior expectations. First, the posterior contrast Δ rises in N under fixed K , not falls. The naive form of the hypothesis would have predicted the opposite. Operationalizing the question as “how much information does an acceptance label carry” (mutual information / tail recall) rather than “how different are accepted and rejected papers on average” is essential for getting the right qualitative picture. Second, linear capacity rescues researcher identifiability *completely* in our simulation ($\rho \approx 0.78$ flat in M_r), a stronger positive result than we expected.

5.1 Limitations

Modeling assumptions. We assume reviewer scores are conditionally independent given θ . Real reviews share author/institution/topic priors, which would inflate effective σ and accelerate the decay we report. Our θ is one-dimensional; real paper quality is multidimensional (novelty, rigor, impact), and our θ should be read as a projection onto a single useful axis. The researcher submission rate S is fixed; in reality, $S(\alpha)$ is increasing in ability [Kwiek and Roszka, 2026], which would partly rescue identifiability under fixed K —softening but not reversing the H3 conclusion.

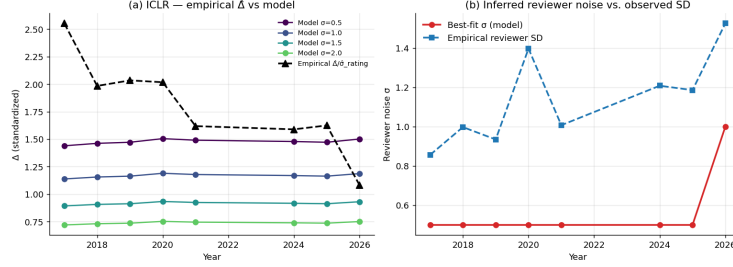


Figure 7: Model calibration to ICLR. With ICLR’s empirical accept rate as the capacity policy, the reviewer noise σ that minimizes RMSE on $\hat{\Delta}/\hat{\sigma}_r$ rises from ~ 0.5 to ~ 1.0 between 2017 and 2026. This is precisely the load-induced-noise channel: linear capacity expansion preserves the analytical signal, but only if the reviewer pool expands too, which it has not.

Empirical reach. The pre-2017 ICLR OPENREVIEW JSON files lack reviewer-rating fields, so we cannot measure dispersion before then. Eight years of usable rating data are enough to estimate a -0.32 ± 0.08 slope but not for a fully saturated time series. Other AI venues are smaller and have shorter rating histories; they are consistent with the pattern but not pinned down by it. The trend results are observational—load-induced noise is the most plausible explanation for the empirical degradation, but coexisting contributors (paper-topic broadening, ML reviewer pool shifts, LLM-assisted reviewing [Latona et al., 2024]) cannot be ruled out.

Reader trust collapse. Our negative-experience feedback rule did not trigger collapse with $\tau_{\text{bad}} = -0.3$ because the average accepted paper has positive quality under both regimes. A more demanding threshold (e.g., expecting top-quartile quality from a prestigious venue) would change the qualitative picture; that case is the natural next experiment.

Generalization. ICLR may not generalize to fields with different review structures (single blind, journal-with-revisions, post-publication review). Our model is venue-agnostic but our empirical anchor is one venue.

5.2 Broader implications

If the -0.32 log-linear decay is real and continues, two practical recommendations follow. **For readers and committees:** the venue label should be treated as one signal among many, not a decisive gate—author identity, topic, and post-publication signals must do the rest of the work. **For venues:** maintaining acceptance rate is not enough. Either the reviewer pool must grow proportionally to submissions (so σ stays constant), or the venue must accept that its certification function is decaying. Mechanism-design changes (rebuttal, AC discussion, transparent OPENREVIEW, structured rubrics) plausibly reduce $\sigma^2(N)$ but their net effect is empirically open.

6 Conclusion

We formalize gate-keeping as a capacity-constrained Bayesian signaling problem at three layers (paper, reader, researcher), parameterized by the submission volume N that prior signaling models hold fixed. Analytically, fixed capacity drives mutual information and tail recall to zero in N (Kendall $\tau \leq -0.95$ uniformly), and fixed-capacity researcher evaluation collapses to bean counting once the field is large enough to make publication count tied at zero for almost everyone (95.5% at $M_r=20,000$). Empirically, 14 years of ICLR data ($N : 67 \rightarrow 19,814$) show a log-linear decay of the standardized posterior contrast at -0.32 per e -fold of N ($R^2 = 0.85$, $p = 1.1 \times 10^{-3}$), accompanied by a $0.86 \rightarrow 1.53$ rise in mean per-paper reviewer SD. The two patterns are jointly consistent with a **load-induced noise** interpretation: ICLR expanded capacity to keep accept rate constant, but the reviewer pool did not expand proportionally, so σ grew and the signal degraded anyway.

Key takeaway. No capacity policy preserves all three of per-paper precision, researcher identifiability, and reader-level relevance as N grows; either the signal degrades or the pool from which the signal selects becomes too large for downstream consumers to use. The venue label survives as a necessary filter and ceases to be a sufficient one.

Future work. Three directions sit naturally next to this paper. (i) An empirical ablation across venue formats (single-blind, double-blind, structured OPENREVIEW, journal-with-revisions) would let us estimate $\sigma^2(N, \text{format})$ and test whether mechanism-design changes can break the load-induced noise channel. (ii) An adaptive capacity policy $K(N) = \alpha \log N$ trades precision and recall in a different way than the three policies we test; the analytical machinery is in hand and the sweep is one parameter. (iii) The reader-trust-collapse model deserves a closed-form treatment with venue priors and a misspecification penalty, so that the threshold at which collapse becomes generic can be characterized in terms of N , σ , and reader expectations rather than a numerical τ_{bad} .

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